

Dohwan Ko

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EDUCATION	Korea University M.S. & Ph.D. integrated in Computer Science and Engineering Advisor: Hyunwoo J. Kim	Seoul, Republic of Korea Mar. 2021 – Feb. 2027 (expected)
	Korea University B.S. in Computer Science and Engineering	Seoul, Republic of Korea Mar. 2017 – Feb. 2021
RESEARCH INTERESTS	• Multi-Modal Agent AI and Reasoning • Video Large Language Models and Foundation Models • Efficient Multi-Modal Video Understanding	
WORKING EXPERIENCE	Meta <i>Research Scientist Intern, Meta Reality Labs</i>	Menlo Park, CA April 2026 – Jul. 2026
	• Enhancing small vision-language models with dual-axis latent reasoning along the sequence and depth axes. • Collaborators: Patrick Huber, Ernie Chang, Adithya Sagar, Chinnadhurai Sankar, Pierce Chuang	
	Meta <i>Research Scientist Intern, Meta Reality Labs</i>	Menlo Park, CA Mar. 2025 – Jun. 2025
	• Developing a streaming video memory agent for query-agnostic online frame selection in online video understanding. • Collaborators: Pierce Chuang, Debojeet Chatterjee, Ashish Shenoy, Yichao Lu, Shane Moon, Luna Dong, Vikas Bhardwaj.	
	NEC Labs America <i>Research Scientist Intern, Media Analytics Team</i>	San Jose, CA Jun. 2024 – Aug. 2024
	• Developing a vision-language model enhanced with kinematic instruction tuning for spatio-temporal reasoning over object speed, movement direction, and traveled distance. • Collaborators: Yumin Suh, Vijay Kumar B G, Manmohan Chandraker.	
PUBLICATIONS	NAVER <i>Engineering Intern, User Feedback Platform</i>	Seongnam, Republic of Korea Jul. 2020 – Aug. 2020
	• Developing a model for malicious comment detection in user-generated content.	
	[1] WearableQA: A Benchmark for Health Reasoning over Real-World Wearable Data Ji Soo Lee, Xilun Chen, Pierce Chuang, Ashish Shenoy, Jason Wei, <u>Dohwan Ko</u>, Hyunwoo J. Kim, Benoit Corda <i>under review</i>. [paper] [2] Retrieval-augmented skill optimization with cross-domain harness retargeting Jaewon Chu, Ji Soo Lee, Jeehye Na, Taehoon Lee, Minseo Yoon, Jihwan Park, Seunghun Lee, <u>Dohwan Ko</u>, Minseok Joo, Yunyang Xiong, Hyunwoo J. Kim <i>under review</i>. [3] SuperClip Pyramid for Video Temporal Grounding Sanghyeok Lee, Juyeon Ko, Joonmyung Choi, Jongha Kim, <u>Dohwan Ko</u>, Ji Soo Lee, Hyunwoo J.	

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under review.

- [4] **VIDEOSEARCH-R1: Iterative Video Retrieval and Reasoning via Soft Query Refinement**
Seohyun Lee*, Seoung Choi*, Dohwan Ko*, Jongha Kim, Hyunwoo J. Kim
European Conference on Computer Vision (ECCV), 2026. [paper]
- [5] **SVMemAgent: A Streaming Video Memory Agent for Query-Agnostic Online Frame Selection**
Dohwan Ko, Ji Soo Lee, Pierce Chuang, Debojeet Chatterjee, Ashish Shenoy, Yichao Lu, Shane Moon, Luna Dong, Vikas Bhardwaj, Hyunwoo J. Kim
Wearable AI Workshop at the European Conference on Computer Vision (ECCVW), 2026. [paper]
- [6] **MoE-GRPO: Optimizing Mixture-of-Experts via Reinforcement Learning in Vision-Language Models**
Dohwan Ko, Jinyoung Park, Seoung Choi, Sanghyeok Lee, Seohyun Lee, Hyunwoo J. Kim
Computer Vision and Pattern Recognition (CVPR), 2026. [paper]
- [7] **DocPrune: Efficient Document Question Answering via Background, Question, and Comprehension-aware Token Pruning**
Joonmyung Choi, Sanghyeok Lee, Jongha Kim, Sehyung Kim, Dohwan Ko, Jihyung Kil, Hyunwoo J. Kim
Computer Vision and Pattern Recognition (CVPR), 2026. [paper]
- [8] **Bidirectional Likelihood Estimation with Multi-Modal Large Language Models for Text-Video Retrieval**
Dohwan Ko*, Ji Soo Lee*, Minhyuk Choi, Zihang Meng, Hyunwoo J. Kim.
International Conference on Computer Vision (ICCV) **Highlight** (top 9.7%), 2025. [paper]
- [9] **ST-VLM: Kinematic Instruction Tuning for Spatio-Temporal Reasoning in Dynamic Videos**
Dohwan Ko*, Sihyeon Kim*, Yumin Suh, Vijay Kumar B G, Minseo Yoon, Manmohan Chandraker, Hyunwoo J. Kim
Preprint, 2025. [paper]
- [10] **LLaMo: Large Language Model-based Molecular Graph Assistant**
Jinyoung Park, Minseong Bae, Dohwan Ko, Hyunwoo J. Kim.
Advances in Neural Information Processing Systems (NeurIPS), 2024. [paper]
- [11] **Large Language Models are Temporal and Causal Reasoners for Video Question Answering**
Dohwan Ko*, Ji Soo Lee*, Wooyoung Kang, Byungseok Roh, Hyunwoo J. Kim.
Empirical Methods in Natural Language Processing (EMNLP), 2023. [paper]
- [12] **Open-Vocabulary Video Question Answering: A New Benchmark for Evaluating the Generalizability of Video Question Answering Models**
Dohwan Ko, Ji Soo Lee, Miso Choi, Jaewon Chu, Jihwan Park, Hyunwoo J. Kim.
International Conference on Computer Vision (ICCV), 2023. [paper]
- [13] **MELTR: Meta Loss Transformer for Learning to Fine-tune Video Foundation Models**
Dohwan Ko*, Joonmyung Choi*, Hyeong Kyu Choi, Kyoung-Woon On, Byungseok Roh, Hyunwoo J. Kim.
Computer Vision and Pattern Recognition (CVPR), 2023. [paper]
- [14] **Randomly Shuffled Convolution for Self-Supervised Representation Learning**
Youngjin Oh*, Minkyu Jeon*, Dohwan Ko, Hyunwoo J. Kim.
Information Sciences, 2022. [paper]
- [15] **Video-Text Representation Learning via Differentiable Weak Temporal Alignment**
Dohwan Ko, Joonmyung Choi, Juyeon Ko, Shinyeong Noh, Kyoung-Woon On, Eun-Sol Kim, Hyunwoo J. Kim.
Computer Vision and Pattern Recognition (CVPR), 2022. [paper]
- [16] **Search-and-Attack: Temporally Sparse Adversarial Perturbations on Videos**

Hwan Heo*, Dohwan Ko*, Jaewon Lee*, Youngjoon Hong, Hyunwoo J. Kim.
IEEE ACCESS, 2021. [paper]

TALKS AND SEMINARS	Transformer and Visual Question Answering Korea University	Nov. 2023
TEACHING EXPERIENCE	Deep Learning (COSE474) Teaching Assistant at Korea University	Seoul, Republic of Korea Sep. 2022 - Dec. 2022
	Artificial Intelligence (COSE361) Teaching Assistant at Korea University	Seoul, Republic of Korea Mar. 2022 - Jun. 2022
HONORS & AWARDS	Outstanding Paper Award , Korea University	2024
	Semester High Honors , Korea University	2020
	Veritas Program Scholarship , Korea University	2017
ACADEMIC SERVICES	Reviewer of ICLR (International Conference on Learning Representations)	2026–2027
	Reviewer of NeurIPS (Advances in Neural Information Processing Systems)	2025–2026
	Reviewer of ICCV (International Conference on Computer Vision)	2025
	Reviewer of CVPR (Computer Vision and Pattern Recognition)	2024–2026
	Reviewer of ECCV (European Conference on Computer Vision)	2024
	Reviewer of WACV (Winter Conference on Applications of Computer Vision)	2027
	Reviewer of ACL ARR (Association for Computational Linguistics Rolling Review)	2024–2025
	Reviewer of AAAI (Association for the Advancement of Artificial Intelligence)	2027
SKILLS	Language • Korean (<i>native</i>) • English (<i>intermediate</i>) Programming C, Python, PyTorch, git, $\LaTeX$	